Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/21/2011
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Mapping gravitropic signaling and response: from ions to growth		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/27/2009	**End Date:**	05/26/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	June 2010: A postdoc has been hired to work on the project, Dr. Won-Gyu Choi, who received his Ph.D. in 2009 from the University of Tennessee. June 2011: An undergraduate, Amanda Miller, is also working with Dr. Choi on the project as a for credit project.		
COI Name (Institution):	Masson, Patrick (University of Wisconsin-Madison) Spalding, Edgar (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX09AK80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research will address program priority 1 in the plant biology element of this NRA by seeking to define the intra-/inter-cellular mechanisms behind the generation and transmission of the gravity signal. It is widely accepted that sedimentation of starch filled amyloplasts in the columella of the root cap and endodermis of the hypocotyl act to initiate the gravitropic signal. However, although pH and Ca2+-based signals have been proposed as key elements in this plant response, the precise mechanisms translating physical force to biochemically transmissible signal is still poorly defined. Needed is a high resolution spatiotemporal map of changes in ion activities induced by the gravity signal. Such mapping will be achieved by confocal imaging of plants expressing a range of GFP-based probes targeted to different subcellular locales within the Arabidopsis seedling. In parallel, we will use machine-vision technologies to quantify the distribution of curvature along the root and hypocotyl axes during gravitropism. Curvature and ion change distribution maps will be measured in wild type and mutants known to be defective in the pathways that lead to gravity signal transduction. By mapping the detailed spatial and temporal dynamics of ion signaling to these morphometric analyses, we will be able to pinpoint where and when each component is altering tropic response. Potential for translation to future spaceflight opportunities: These studies will develop a morphometric technology highly applicable to spaceflight analysis where existing hardware can be used to collect the time-lapse imaging required for morphometric analysis. Such analysis will develop a quantitative description of the effects of the flight environment on plant development at levels of architecture ranging from branching patterns to the kinematics of growth. This will allow us to ask questions only addressable by the unique environment of Space, for example defining the effects of mutations on gravitational response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has capitalized on the development of automated image analyses algorithms that allow for the automated extraction of quantitative parameters of growth such as growth rate, organ angle, rate of curvature. Although these are being applied to root gravitropism in Arabidopsis, they hold the potential to be applicable to a much wider set of growth responses, allowing for the rapid screening of plants for quantitative differences in their growth patterns. The work is also developing the use of calcium-sensitive green fluorescent protein-based probes that have very high affinity for calcium. These are allowing for the visualization of calcium signals that were previously undetectable and has applications to a wide range of stress signaling pathways in plants.
Task Progress:	This grant covers research to map the dynamics of ionic signaling associated with gravity sensing and response onto the kinetics of gravitropic growth. This analysis will be used to assess how well such signaling events can be linked to the control of plant development. Over the last year we have concentrated on: (1) completing a catalog of morphometric analysis of gravitropic growth responses from the root systems of a range of gravity and growth response mutants in Arabidopsis and (2) simultaneously imaging ionic signaling events in wild type plants and in these mutant backgrounds (3) developing high-resolution kinematic analyses of cell-level growth in these plants and (4) characterizing the effects of medium composition and especially potential anoxic effects on these responses. (1) Defining gravitropic growth kinetics in wild type and gravitropism-related mutants. We have completed a high-resolution morphometric analysis of gravitropic response in wild-type Arabidopsis in a range of ecotypes to define the natural variability in “wild-type” response and also analyzed a range of mutants to define precisely where in time and space in their gravitropic response they exhibit alteration in growth control. We have analyses of mutants including pgm1 (which has a lesion in phosphogucomutase and so is unable to lay down starch in the gravity sensing cells), aux1 (which has a lesion in an auxin influx transporter that disrupts auxin-related events associated with gravity responses), arg1 (lesion in a DNAJ protein that acts in the gravity sensory cells of the root cap) and mar1 and mar2 both alone and in the arg1 background (MAR1 and 2 are proteins found in the plastid membrane possibly playing a role in gravity sensing). Interestingly, and for unknown reasons at present, mar1 and mar2 mutants only exhibit effects on gravitropic growth response when a mutation in arg1 is also present. However, we have now characterized that alone mar1 exhibits a reversal in gravitropic response in the root which is dependent on light level. This appears to represent an interaction of gravitropic and phototropic response. Current work is aimed at defining the precise light dependency with regard to wavelength and possible involvement of phototropic receptors such as the phototropins (blue light) or phytochromes (red light sensors). We have also tested response to different growth conditions, concentrating on sucrose supplementation to the medium. This analysis has allowed us to define that the mutants in phosphogucomutase that are characterized as showing reduced gravitropism (thought to be due to a defect in laying down starch required for perception in the gravity sensing cells) actually show a conditional phenotype with higher levels of sucrose progressively suppressing the gravitropic phenotype and moving more and more to wild-type-like response kinetics. This rescue saturates around 5 mM sucrose. This level of sucrose is much less than the 30 mM usually added to assay media. We speculate that this recovery of gravitropic response is linked to an alternate pathway of starch deposition that bypasses the phosphoglucomutase-dependent step and that is facilitated by increased sucrose levels. We are currently verifying this sucrose-dependent starch deposition. If this hypothesis is correct it could help explain the residual gravitropic response seen in many experiments with these phosphoglucomutase mutants. This residual gravitropic response remains one of the observations that is proposed to contradict the predictions of the most widely held model for plant gravity perception, the starch statolith hypothesis where starch in the plastids is required to elicit graviperception. (2) Developing cellular imaging approaches to capture early signaling events in the gravitropic response. We have transformed the GFP-based Ca2+ imaging probe cameleon YC3.6 into the mutant backgrounds described above and into 8 different “wild-type” ecotypes of Arabidopsis for which we have morphometric growth data and are currently imaging the Ca2+ changes occurring upon gravistimulation (applied by rotation of the plant through 90 degrees) using confocal imaging. We have also investigated whether measurements would be more readily made using the alternate Ca2+ imaging probe aequorin. Aequorin is a luminescent protein that is responsive to Ca2+ levels, and has advantages of being less sensitive to artifacts related to shifts in imaging plane that occur as gravistimulation is applied than GFP-based probes. We have therefore characterized plants stably transformed with this probe driven by a ubiquitously expressed promoter (the 35S promoter). Our analysis of the aequorin plants suggests signal strength is severely limiting for detection of Ca2+ changes to all but the largest, most widespread Ca2+ signals such as generated by cold-shock, mechanical stimulation or chemical elicitors such as extracellular ATP and does not resolve Ca2+ changes elicited by gravitropic stimulation. We therefore optimized the confocal imaging of the YC3.6 lines to be able to resolve rapid changes in Ca2+ levels in plants exhibiting gravitropic signaling/response and to minimize any focal plane shifts that occur as gravistimulation is applied. Our analysis suggests multiple Ca2+ signals separated in time occur in the root cap

upon this stimulation. The major Ca2+ change occurs 10 minutes after gravistimulation, is asymmetrical and limited to one or two cells in the lateral root cap on the lower side of the root. These cells exhibit complex Ca2+ oscillations. The delayed response of these cells suggests the changes play a role in signal export, possibly related to auxin transport in the root cap and coincide with the first detectable growth responses of the root as defined by our morphometric and kinematic analysis. However, detailed analysis of the kinetics of auxin response reporters such as the DR5 promoter indicate these are detectably activated more than 2 hours after these signals have occurred.

As part of the development of this Ca2+ imaging technology, we have been using the addition of extracellular ATP as a positive control. Extracellular ATP is an emerging regulator of plant growth, with mounting evidence for a role in gravitropic response. In collaboration with the laboratories of Gary Stacey (University of Missouri, Columbia) and Alan Jones (UNC) we have characterized an oscillatory Ca2+ response to this ATP treatment that appears dependent on signaling through a G-protein signal transduction element. This characterization has been published in Plant Physiology. We have also introduced the YC3.6 sensor into mutants of the G-protein complex to assess the cell-type specific nature of these signaling systems.

We have also been collaborating with the lab of Janet Braam (Rice University) on imaging Ca2+ signaling in calmodulin-like protein mutants that likely form part of the response network for Ca2+ signaling changes. We have determined that one such mutant (cml24) has effects on root growth, leading to root skewing, cell file rotation and altered microtubule orientation, suggesting a model whereby Ca2+ and CML24 act to control microtubule function and so contribute to the control of root directional growth. This work has been accepted for publication in the Journal of Plant Growth Regulation.

3) Role of anoxia on root gravitropic growth responses. During our analysis of gravitropic growth, we have become increasingly aware that root zone hypoxia/anoxia alters the kinetics of the response. In addition, we have also characterized, using our YC3.6-expressig plants, that anoxia due to poor oxygenation of growth media leads to a complex Ca2+ signaling response in the root tip. Our mining of the publicly available transcriptional databases suggests that Ca2+ response proteins such as the calcium-dependent protein kinases, calmodulin-like proteins and Ca2+ transporters such as the CAX family are highly represented in datasets from anoxia-treated plants. At a practical level for experimentation, anoxic response may therefore act to both trigger Ca2+ changes in the root that may confound analysis of gravitropic signaling response and also rewire the Ca2+ response machinery to gravitropic Ca2+ signals. In addition, a major limitation to root growth in spaceflight is thought to be root zone hypoxia/anoxia.

Bibliography Type:	Description: (Last Updated: 02/02/2026)
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Books/Book Chapters	Choi W-G, Swanson SJ, Gilroy S. "Calcium, mechanical signaling and tip growth." in "Coding and Decoding of Calcium Signals in Plants." Ed. S. Luan. New York, NY : Springer, in press as of June 2011. Expected publication August 2011., Jun-2011